Lecture Notes 3
More on Growth Theory

Intermediate Macroeconomics, EC2211

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Fall 2025

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Contents and literature

- The Solow model with population growth
- Convergence
- Human capital
- Technological progress
- Finite natural resources
- Malthus and the dismal science

Literature:

- Jones (2024), chapter 5 and 7.7-7.8
- *Steinsson (2025a), "Malthus and pre-industrial stagnation"

The Solow model with population growth

The Solow model with population growth

- Recall the Solow model from the previous lecture
- Let us now allow for population growth in this model

  Let L_t denote the size of the population at date t

The production function

The production function is

$$Y_t = AK_t^{\alpha} L_t^{1-\alpha} \tag{1}$$

As before, we let lower-case letters denote variables in per capita terms, i.e. $k_t \equiv K_t / L_t$ and

$$y_t \equiv \frac{Y_t}{L_t} = \frac{AK_t^{\alpha} L_t^{1-\alpha}}{L_t} = Ak_t^{\alpha} \tag{2}$$

Population growth

Let n be the (constant) population growth rate, so that

$$\frac{L_{t+1}}{L_t} = 1 + n$$

Steady state with population growth

Recall that the capital stock develops according to

$$K_{t+1} = (1-\delta)K_t + sY_t \tag{3}$$

Divide by L_t:

$$\frac{K_{t+1}}{L_t} = (1-\delta)\frac{K_t}{L_t} + s\frac{Y_t}{L_t} = (1-\delta)k_t + sy_t$$

Multiply the left-hand side by L_{t+1}/L_{t+1}:

$$\frac{K_{t+1}}{L_t} = \frac{L_{t+1}}{L_{t+1}}\frac{K_{t+1}}{L_t} = \frac{L_{t+1}}{L_t}\frac{K_{t+1}}{L_{t+1}} = (1+n)k_{t+1}$$

Since $nk_{t+1} \approx nk_t$ we obtain:

$$(1+n)k_{t+1} = (1-\delta)k_t + sy_t \implies \Delta k_{t+1} \approx sAk_t^{\alpha} - (\delta+n)k_t \tag{4}$$

Steady state with population growth

Steady state when $k_{t+1} = k_t$. Equation (4) implies that k^* satisfies:

$$sA(k^*)^\alpha = (\delta + n)k^*$$

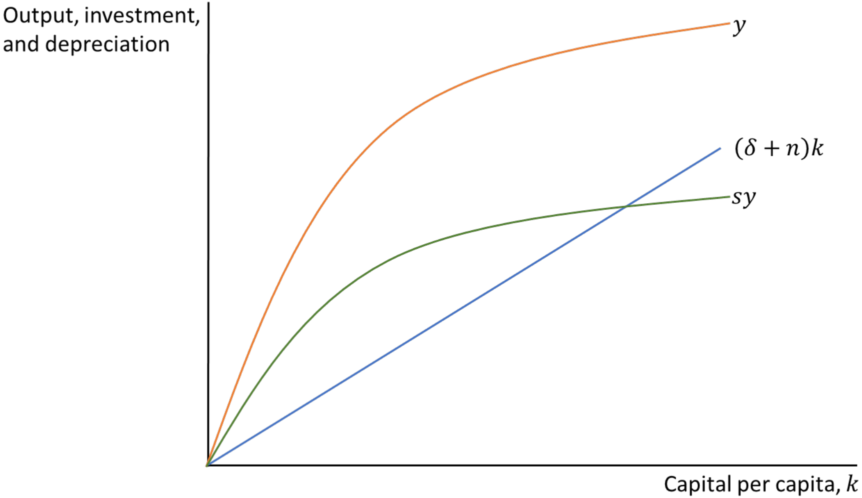
Solving for k^*:

$$k^* = \left(\frac{sA}{\delta + n}\right)^{\frac{1}{1-\alpha}} \tag{5}$$

Using $y = Ak^\alpha$, we obtain y^*:

$$y^* = \left(\frac{sA}{\delta + n}\right)^{\frac{\alpha}{1-\alpha}} \tag{6}$$

The Solow diagram with population growth



Steady state with population growth

Note how the steady state is affected by population growth:

- Consider a fixed amount of investment, sY
- With higher population growth, each child will have less capital
  - So higher n means that we move to a steady state with less capital per person
  - Alternatively, we need a higher savings rate to maintain the same capital stock per person over time

Growth rates on the balanced growth path

Balanced growth path (BGP)

On a balance growth path, all variables grow at constant rates (or are constant)

Variable	Notation	Growth rate on BGP
Capital per worker	$k = K/L$	0
Output per worker	$y = Y/L$	0
Capital stock	$K = k \cdot L$	n
Output	$Y = y \cdot L$	n

Note: I will often use "steady state" as a synonym for "balanced growth path"

The basic Solow model: summing up

- An important insight from the Solow model:
  - **Increased capital accumulation does not lead to higher growth in the long run**
  - An increase in the savings rate (more investments) generates growth during the transition to a new equilibrium
  - In the new equilibrium, output is higher but the growth rate is the same
- The diminishing marginal product of capital is key:
  - If we invest and build more capital, output will increase but so will depreciation of capital
  - Depreciation is linear in capital but output increases less and less as we get more capital
- This conclusion points to the importance of TFP: technological progress is necessary for long-run growth

Convergence (again)

Convergence and conditional convergence

Consider two economies, A and B. They are similar, except that population growth is higher in B than in A

How will the two countries differ in:

- steady-state growth in output per capita?
- steady-state level of output per capita?
- transitional growth rates in output per capita if they initially have equally large capital stocks per capita?

Convergence and conditional convergence

Rewrite equation (4) to see that the growth rate of k is

$$\frac{\Delta k_{t+1}}{k_t} = s\frac{y_t}{k_t} - (\delta + n)$$

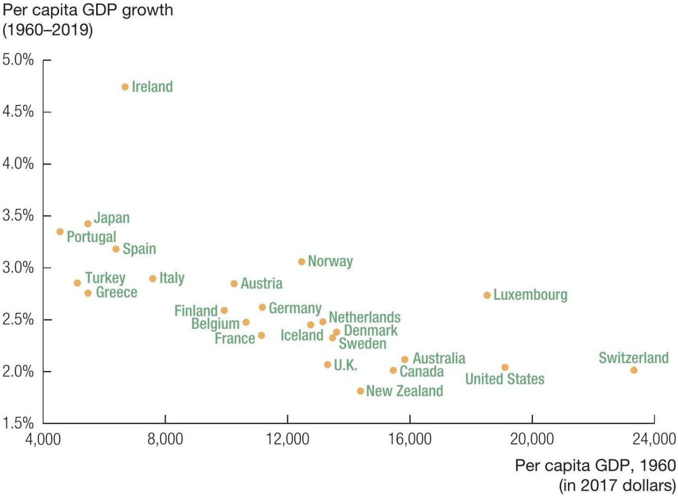
The growth rate of output per capita is (show this!)

$$\frac{\Delta y_{t+1}}{y_t} = \alpha\frac{\Delta k_{t+1}}{k_t}$$

So if countries A and B have the same k_t and are identical in all other ways except that n is larger in B than in A, we now see that the growth rate will be lower in B than in A

More generally, the Solow model says that convergence is *conditional*: we must account for differences not only in initial capital, but also in population growth, investment rates, etc

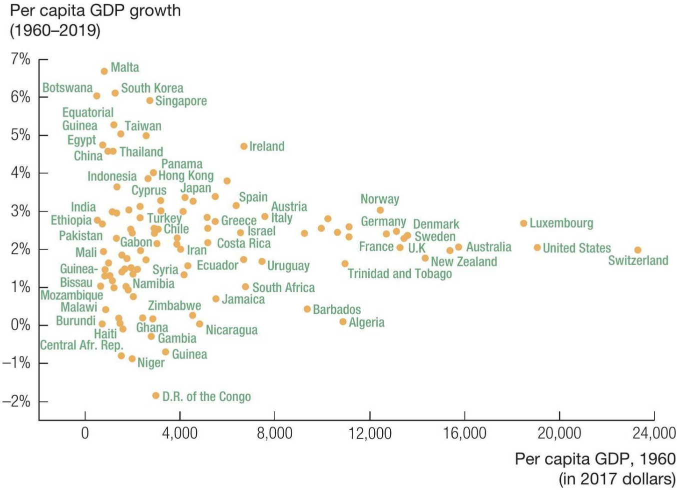
Growth rates in the OECD, 1960-2019



Source: Penn World Table, Version 10.0.

Figure 5.8 in Jones (2024)

Growth rates around the world, 1960-2019



Source: Penn World Table, Version 10.0.

Figure 5.9 in Jones (2024)

Middle income traps?

Patel, Sandefur and Subramanian (2021):

- "Poorer countries have in fact been catching up with richer ones, albeit slowly, since the mid 1990s"
- "Debates about a 'middle-income trap' also appear anachronistic"

World Bank (2024):

- "The progress of the middle-income countries has slowed in recent decades"
- "A key characteristic of economic growth in middle-income countries is its lack of persistence"

Growth rates around the world, 1960-2019

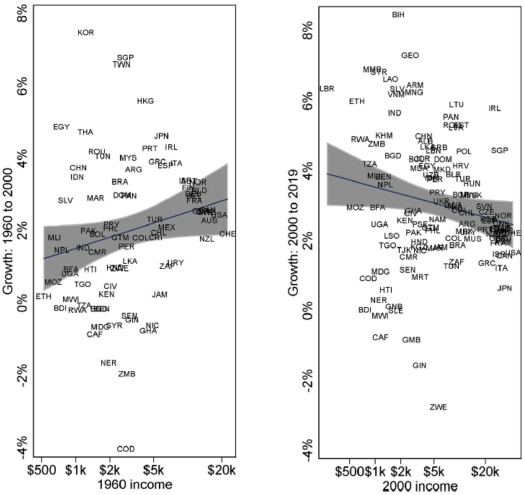


Fig. 3. Changing Patterns of Convergence: Two Non-Overlapping Periods. Note: The horizontal axis shows the natural log of real per capita GDP from the Penn World Tables version 10.0., and the vertical axis shows average annual real growth over the period listed. Shaded area represents a 95% confidence interval around the regression line.

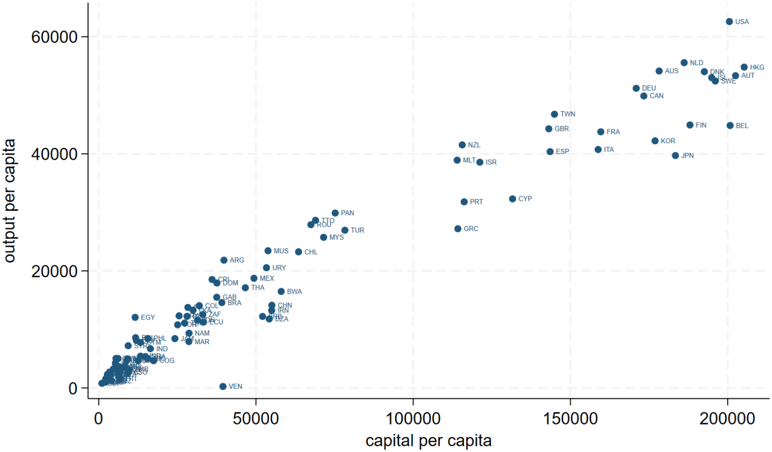
From Patel, Sandefur and Subramanian (2021)

Human capital

Mankiw, Romer & Weil

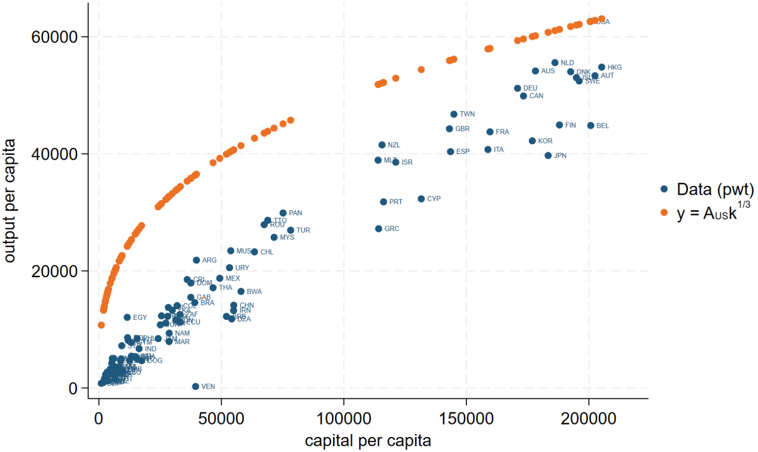
- Recall that α is the capital share in production, $y = Ak^{\alpha}$
- With $\alpha = 1/3$, the production function predicts "too high" output in countries with little capital
- The fit between k and y appears better when $\alpha = 2/3$

Output in the data



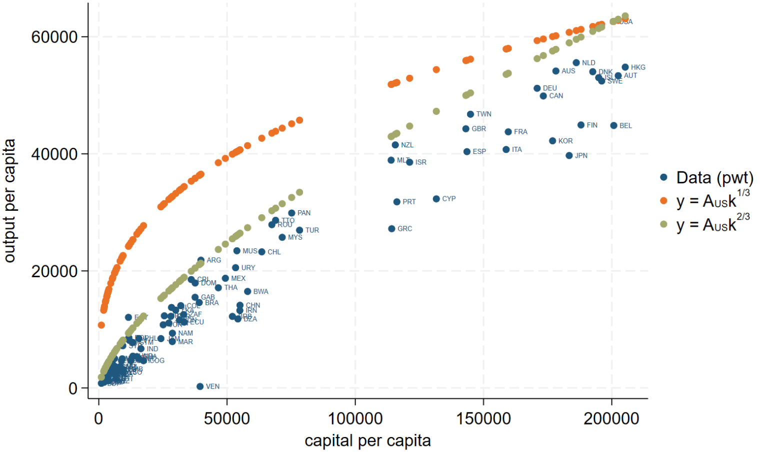
Output and capital per capita 2019, 2017 USD price level. Capital stocks calculated as in Jones (2024). Source: PWT 10.01.

Output in the data and implied by $y = Ak^{1/3}$



Output and capital per capita 2019, 2017 USD price level. Capital stocks calculated as in Jones (2024). Source: PWT 10.01.

Output in the data and implied by $y = Ak^{1/3}$ and $y = Ak^{2/3}$



Output and capital per capita 2019, 2017 USD price level. Capital stocks calculated as in Jones (2024). Source: PWT 10.01.

Mankiw, Romer & Weil

- The fit between k and y appears better when $\alpha = 2/3$
- But if $\alpha = 2/3$, the capital share in production becomes unrealistically high and the wage share much too low
- Is there something missing? Something that is similar to capital but that is compensated as labor?

The importance of human capital

Mankiw, Romer and Weil (1992) modify the Solow model by adding human capital to equation (1):

$$Y_t = AK_t^{\alpha} H_t^{\beta} L_t^{1-\alpha-\beta} \tag{7}$$

where H_t is the stock of human capital and $\alpha + \beta < 1$

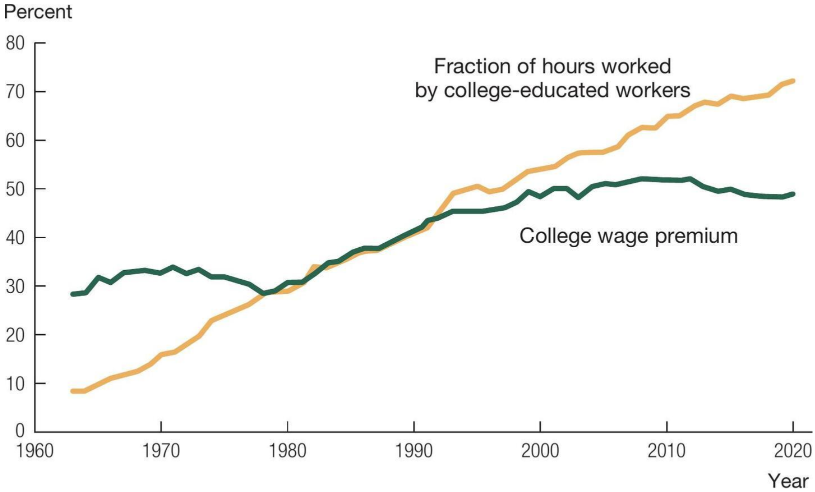
As in the basic Solow model, they assume that a constant fraction (s_k) of output is invested in physical capital. They also assume that a constant fraction (s_h) of output is invested in human capital.

In this model, investment in physical capital increases output, which increases investment in human capital. The effect of investment on income per capita is therefore greater when human capital is taken into account.

Mankiw, Romer and Weil (1992): conclusions

- Mankiw, Romer and Weil take the model with human capital to the data and report highly significant elasticities of income per capita with respect to investment that are consistent with each of α and β around $1/3$
- Moreover, the model with human capital can explain almost 80 percent of the cross-country variation in income per capita in their samples

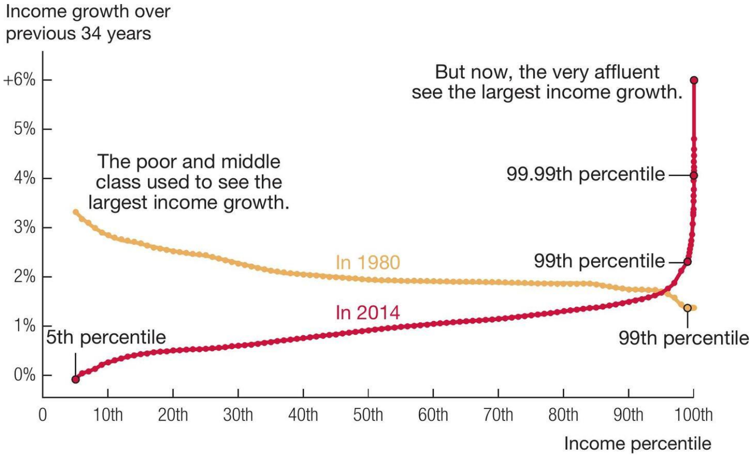
Rising return to education



Source: Daron Acemoglu and Carlos Molina, private communication, April 2, 2023. The college wage premium is computed after controlling for various demographic characteristics such as region, age, race, gender, and experience.

Figure 7.8 in Jones (2024)

Growth and inequality



Source: This figure is based on data from Thomas Piketty, Emmanuel Saez, and Gabriel Zucman, "Distributional National Accounts: Methods and Estimates for the United States," *Quarterly Journal of Economics*, vol. 133 (2018), pp. 553–609. David Leonhardt in "Our Broken Economy, in One Simple Chart," *New York Times*, August 7, 2017, worked with the original authors to create this version that includes the 1980 line.

Technological progress in the Solow model

Technological progress in the Solow model

- We saw in the previous lecture that growth in TFP has been an important factor behind increased output in many countries
- The Solow model is often specified with TFP, A, not as a constant but with the assumption that it grows at an exogenous rate
- The Solow model can then "allow" for long-run growth
  - ... but it does not attempt to explain or model why TFP grows
  - Next time we will look at a model that makes technological progress endogenous

The balanced growth path with TFP growth in the Solow model

- Suppose that the production function is

$$Y_t = A_t K_t^{\alpha} L_t^{1-\alpha}$$

where TFP grows at the constant rate g,

$$A_{t+1} = (1+g)A_t$$

- Can there then be a balanced growth path in this economy?

The balanced growth path with TFP growth in the Solow model

- Recall that along a BGP, all variables grow at constant rates
- Recall from (3) that the capital stock develops according to $K_{t+1} = (1-\delta)K_t + sY_t$, i.e.

$$\frac{K_{t+1} - K_t}{K_t} = s\frac{Y_t}{K_t} - \delta$$

- The left-hand side is the growth rate of capital. The right-hand side can only be constant if Y_t/K_t is constant!
- And for Y_t/K_t to be constant, Y and K must grow at the same rate; call this rate γ

The balanced growth path with TFP growth in the Solow model

- The growth rate of Y is

$$g_Y = g_A + \alpha g_K + (1-\alpha) g_L \tag{8}$$

where g_X denotes the growth rate of variable X

- We have assumed that the growth rate of TFP is $g_A = g$ and that the growth rate of the population is $g_L = n$
- We also noted that along a BGP we must have $g_Y = g_K = \gamma$
- So we can rewrite (8) as $\gamma = g + \alpha\gamma + (1-\alpha)n$, i.e.

$$\gamma = n + \frac{1}{1-\alpha} g$$

The balanced growth path with TFP growth in the Solow model

- What are then the per capita growth rates?
- Output per capita, capital per capita and consumption per capita grow at rate $\gamma - n = g/(1-\alpha)$
- Using similar steps as in our basic Solow model, we could have demonstrated that the economy will move towards the balanced growth path if it starts with any positive level of capital

Taking stock of the Solow model

Taking stock of the Solow model

- Demonstrates how the level of GDP per capita in the long run increases with:
  - high TFP
  - high savings rate (= high investment rate)
  - low depreciation rate
  - low population growth rate
- Demonstrates that diminishing returns imply that capital accumulation cannot generate sustained growth
- Helps us understand growth rates out of steady state
  - "Transitional dynamics"
- Shortcomings:
  - Suggests that TFP differences are key to explaining cross-country differences in income, but TFP differences are exogenous and not explained
  - Does not explain why investment rates vary across countries

Finite natural resources

Finite natural resources

- We have assumed that capital depreciates at an exogenous rate, but that it can be rebuilt, and expanded, through investment
- What if some production factors are not renewable ("oil") and some others are in fixed supply ("arable land")?
- What if the economy affects the climate, and the climate affects the economy? *Not addressed in this course, but very active research field last decade. The Solow model is an important building block for such analysis.*

Non-renewable natural resources

The production function is

$$Y_t = A_t K_t^{\alpha} (u_t R_t)^{\beta} L_t^{1-\alpha-\beta} \tag{9}$$

where R is a nonrenewable natural resource and u_t is the fraction of that resource that is used for production in period t. We also assume that $\alpha + \beta < 1$.

The stock of the natural resource develops according to $R_{t+1} = R_t - u_t R_t$. To simplify, assume that u is constant. The stock of the resource then follows:

$$R_{t+1} = (1-u) R_t$$

As before:

$$K_{t+1} = (1-\delta) K_t + s Y_t \tag{10}$$

Non-renewable natural resources

- Can we find an equilibrium in this economy?
- Can there be positive growth in the long run if a production factor is nonrenewable?

- Equilibrium here would be a balanced growth path where all variables grow at constant rates (rates that may be positive, zero or negative)

Equilibrium with non-renewable natural resources

Note that, as before, the growth rate of capital is (see (10))

$$\frac{K_{t+1} - K_t}{K_t} = s\frac{Y_t}{K_t} - \delta$$

and that this growth rate can only be constant if Y_t and K_t grow at the same rate

We know that A grows at rate g, R at rate $-u$ and L at rate n. Is it then possible for Y and K to grow at a common rate (call this rate γ)?

Equilibrium with non-renewable natural resources

From the production function (9) we see that

$$g_Y = g_A + \alpha g_K + \beta g_R + (1 - \alpha - \beta) g_L$$

Now use $g_Y = g_K = \gamma$, $g_A = g$, $g_R = -u$ and $g_L = n$ to get

$$\gamma = g + \alpha\gamma - \beta u + (1 - \alpha - \beta) n$$

which simplifies to

$$\gamma = n + \frac{1}{1 - \alpha} g - \frac{\beta}{1 - \alpha}(n + u)$$

Interpreting the equilibrium with non-renewable natural resources

We are interested in the growth rate of output per capita, $\gamma - n$:

$$\gamma - n = \frac{1}{1-\alpha} g - \frac{\beta}{1-\alpha}(n+u)$$

If $\beta = 0$, we are back to the original Solow model

If $\beta > 0$, the nonrenewable resource is a drag on growth:

- Rapid population growth $\rightarrow$ more people to share the vanishing resource
- High utilization $\rightarrow$ rapid depletion

Growth can be positive or negative in equilibrium depending on parameters

Land

The production function is now:

$$Y_t = A_t K_t^{\alpha} D^{\lambda} L_t^{1-\alpha-\lambda}$$

where D denotes land. As before, we assume that $\alpha + \lambda < 1$

The setup is very similar to the previous analysis, but D is now a production factor that is constant

Solving for equilibrium we find that the growth rate of output per capita is

$$\frac{\Delta y_{t+1}}{y_t} = \frac{1}{1-\alpha} g - \frac{\lambda}{1-\alpha} n$$

Estimates indicate that the drag from R and D in production on the annual U.S. growth rate is around 0.3 percentage points (see Nordhaus (1992) and Jones and Vollrath (2024))

Malthus and the dismal science

Malthus and the dismal science

- Malthus (1798), "An Essay on the Principle of Population"
- Income growth $\rightarrow$ population growth $\rightarrow$ land shortage (decreasing marginal product of labor) $\rightarrow$ falling income
- Income will be stuck at low level

Thomas Malthus 1766-1834

Malthus and the dismal science

The Malthusian production function:

$$Y_t = AD^{\lambda} L_t^{1-\lambda} \tag{11}$$

where D is again land which is in fixed supply

Population growth is now *endogenous*,

$$\frac{L_{t+1}}{L_t} = \frac{y_t}{y^s} \tag{12}$$

Population thus grows when income per capita is high. But the population *shrinks* if income per capita is below the "subsistence level" y^s

Solving the Malthusian model

Rewrite equation (11) in per capita terms:

$$y_t = A\left(\frac{D}{L_t}\right)^{\lambda} \tag{13}$$

and use this in (12):

$$L_{t+1} = \frac{AD^{\lambda}}{y^s} L_t^{1-\lambda} \tag{14}$$

Let

$$\phi = \frac{AD^{\lambda}}{y^s}$$

so that we get

$$L_{t+1} = \phi L_t^{1-\lambda}$$

See figures 3 and 4 in Steinsson (note that his N is the same as our L and his a is our λ)

Interpreting the Malthusian model

Consider the effects of an increase in productivity, A:

- Production and wages increase
- The $L_{t+1} = \phi L_t^{1-\lambda}$ curve shifts up
- The population grows until it reaches a new equilibrium
- In the new equilibrium, production per capita and wages are back at the original level

Fixed supply of land in combination with diminishing marginal product of labor is key here:

- Higher production leads to a larger population
- In contrast to the Solow model where we can build capital, here land is fixed
- Output increases, but less and less as land gets crowded – so output per capita falls
- Lower incomes per capita means that the population shrinks

The Malthusian model with technological progress

Suppose that productivity grows over time at the constant rate g,
$A_{t+1} = (1+g)A_t$

Will per capita income now grow in the long run?

Consider the two key equations (13) and (14) in the Malthusian model, with A_t instead of A:

- Income per capita is

$$y_t = A_t \left(\frac{D}{L_t} \right)^{\lambda} \tag{15}$$

- and the size of the population develops according to

$$L_{t+1} = \frac{A_t D^{\lambda}}{y^s} L_t^{1-\lambda} \tag{16}$$

The Malthusian model with technological progress

From (15) and (16), we see that the growth rates for y and L (along a balanced growth path where growth rates are constant) are:

$$g_y = g_A - \lambda g_L \tag{17}$$

and

$$g_L = g_A + (1 - \lambda) g_L \tag{18}$$

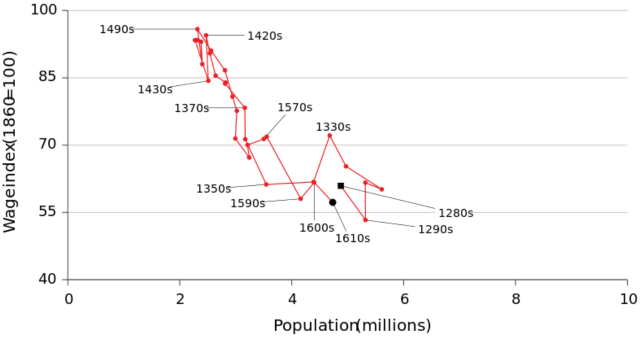
Combining these two equations, and using $g_A = g$, we get:

$g_L = g/\lambda$

$g_y = 0$

According to the Malthusian model, productivity growth (in the long run) only generates higher population growth, not higher income!

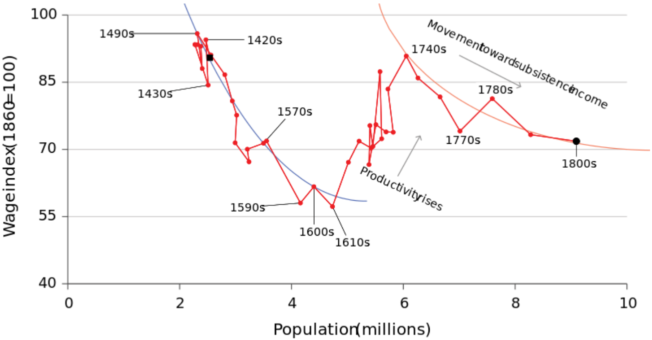
The Malthusian model: evidence



- Population fell in the 1300s due to the plague; wages then rose
- As wages rose, the population increased but then wages fell
- In 1610, population and wages were close to the levels in 1280

Material from COREecon: The Economy 2.0

The Malthusian model: evidence



- Technological progress in the 1600s (crop rotation, potatoes, Chinese plough)
- Wages initially increased
- But eventually wages fell back as the population rose

Material from COREecon: The Economy 2.0

Malthus' unfortunate timing

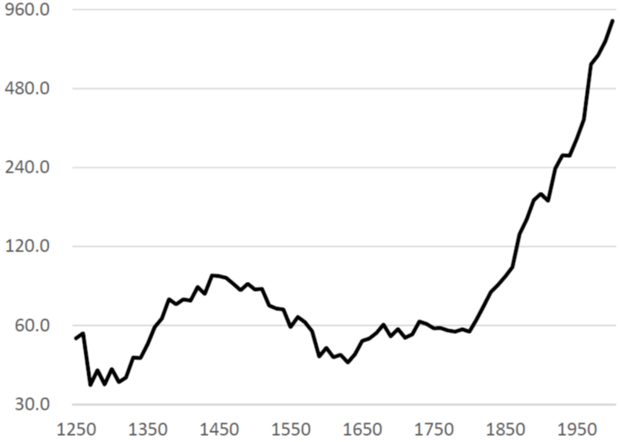


Figure 1: Real Wages of Laborers in England from 1250 to 2000

Figure 1 in Steinsson

From Malthus to Solow

Why did the world economies eventually escape the Malthusian trap?

- Very rapid increase in productivity growth, population increased but "not enough" to reduce income
- Land became a less important production factor, physical capital, machines, etc became more important
- Demographic transition (at high income levels, additional income leads to *lower*, not higher, birth rates, ...)

What we did

- The Solow model
  - Population growth
  - Convergence (are poor countries catching up with the frontier?)
  - Human capital (do differences in schooling explain differences in income levels?)
  - Technological progress (the only source of long-run growth in per capita income in the Solow model)
- Finite natural resources
  - There can be long-run growth even if a production function is not renewable, provided that technological progress is sufficiently rapid
- The Malthusian model
  - No long-run growth even if there is technological progress!

Next time

TFP and long-term economic growth

- The AK model
- The Romer model
- Institutions and growth
- Accounting for growth in the United States and in the European Union